Skoltech

MIMO Systems in Wireless Communications

HW- 2

Master's Educational Program: IoT and Wireless Communication

Student: Muhammad Hanzala Iqbal

CONTENTS

Results and Discussion:

Questions:

1. **Discuss what are the properties of observed PDP? How many samples and in which positions you need to describe this signal uniquely? What is the difference for channel link_chan_PATH.mat?**

Answers:

The Power Delay Profile (PDP) describes how the received signal power is distributed across different delay bins of the channel impulse response. From the obtained PDP we can observe several important properties.

First, the majority of the channel energy is concentrated in a **small number of delay bins near the beginning of the delay axis**. The power rapidly decreases after the first few taps, which indicates that the strongest multipath components arrive with small delays. This means the channel has limited effective delay spread, even though the IDFT produces 600 delay samples.

Second, the PDP shows that most of the useful energy can be captured using only a small subset of delay bins. This is confirmed by the calculated energy window metrics:

Channel	**RMS Delay Spread (bins)**	**L90 (90% energy)**	**L99 (99% energy)**
link_chan_2	**77.70**	**9**	**17**
link_chan_PATH	**74.17**	**5**	**22**

These values indicate that:

- For link_chan_2, about 90% of the total channel energy is contained within the first 9 delay bins, and 99% within 17 bins.
- For link_chan_PATH, 90% of the energy is concentrated even more strongly, within only 5 bins, but the remaining weaker multipath components extend further, since 99% of the energy requires 22 bins.

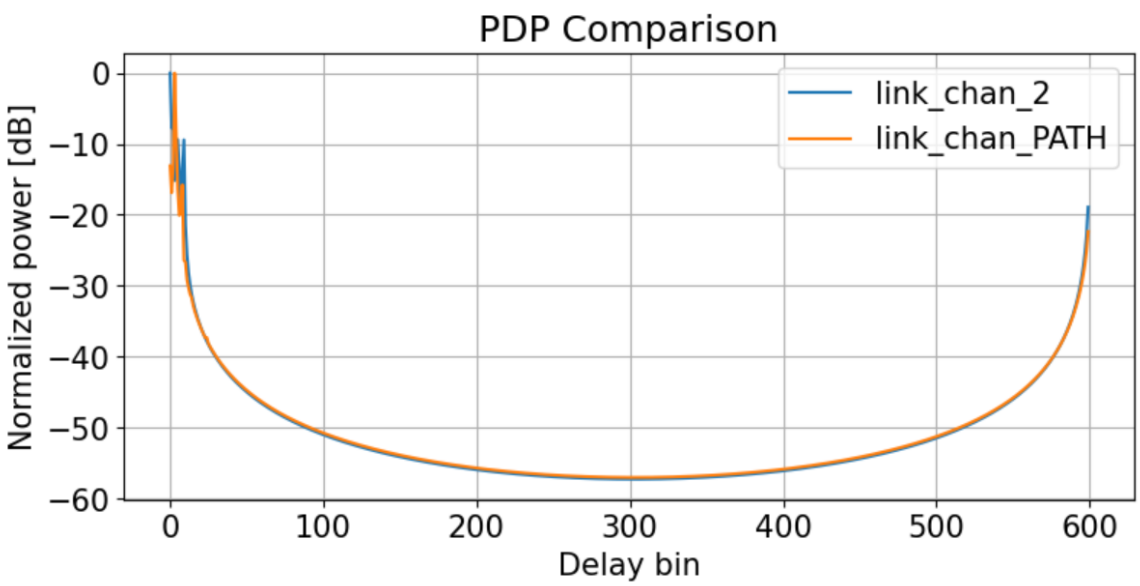
2. Why do we have some non-zero samples closer to the end of PDP

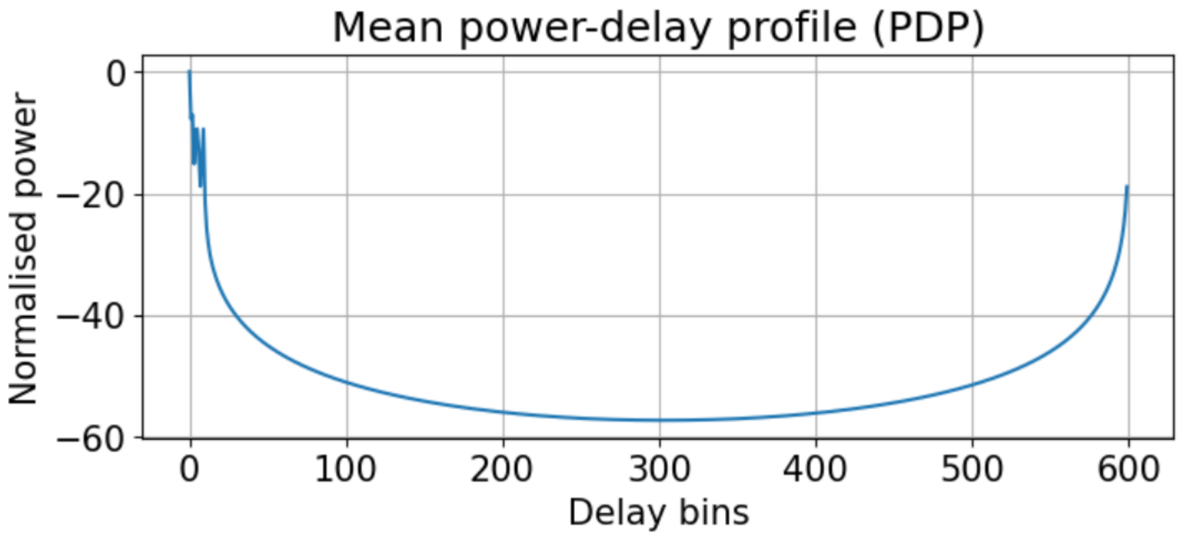
The non-zero samples near the end of the PDP (bins ≈ 550–600) are an artifact of the circular/periodic nature of the IDFT/IFFT used to compute the channel impulse response.

In both channels (link_chan_2 and link_chan_PATH), the physical delay spread is very short most energy arrives with small delays near $\tau \approx 0$. Due to the periodicity of the transform (period K=600), these short-delay paths wrap around and reappear near the end of the delay axis.

The PDP comparison plot shows nearly identical rises at both ends and a very low middle region (≈ –60 dB), confirming no real long-delay paths exist the end-region power is simply the same early paths folded back by the circular IDFT.

This wrap-around effect is expected for channels with limited dispersion and explains why adaptive windows often start near position 540–550 to cover both ends efficiently.





Analysis of Channel Estimation and Precoding Results

2. Channel estimation with noisy pilots and PDP analysis

The channel was estimated using the received pilot signal corrupted by additive white Gaussian noise and a cyclic shift of the Zadoff–Chu sequence equal to **shift = 50**. After correlating the received pilots with the known sequence, a least-squares (LS) estimate of the channel was obtained. The estimated channel was then transformed to the delay domain using the inverse DFT to analyze the power delay profile (PDP).

In the absence of noise, the delay-domain channel impulse response would show a small number of dominant taps corresponding to the physical multipath components. However, when noise is present, it affects the delay-domain representation in several ways.

- First, noise raises the noise floor across all delay bins, making the PDP appear less sparse. Instead of having only a few significant taps, small non-zero values appear across many delay bins.
- Second, noise makes it harder to distinguish weak multipath components from noise, which may lead to inaccurate identification of the true channel support.
- Third, the estimated delay spread may **appear larger than the true physical delay spread,** because noise contributes additional energy across many bins.

Therefore, without additional processing, the LS estimate contains both useful channel taps and noise components distributed across the delay axis.

3. Adaptive Window Selection Based on CIR Power and SNR

To determine appropriate delay-domain window parameters, an adaptive method based on the normalized channel impulse response (CIR) power and the signal-to-noise ratio (SNR) is used.

First, the algorithm analyzes the delay-domain CIR power to identify the strongest propagation path, which corresponds to the delay bin with the maximum power value. This peak indicates the main arrival path of the signal and serves as a reference point for locating the dominant multipath cluster.

Next, the noise floor of the CIR power is estimated using the median value of the delay-domain power. The median is chosen because it provides a robust estimate of the background noise level and is less influenced by strong multipath peaks compared to the mean.

A threshold is then defined by multiplying the noise floor with a scaling factor α, which depends on the SNR. The scaling factor is designed to adapt to channel conditions:

- At high SNR, $\alpha$is small, allowing more delay bins to be considered part of the multipath cluster.
- At low SNR, αincreases, resulting in a stricter threshold that excludes noise-dominated bins.

All delay bins whose power exceeds this threshold are selected as potential components of the main multipath cluster.

To ensure that only the relevant cluster around the dominant path is considered, the algorithm computes the circular distance between these bins and the peak location. Only bins within a certain radius around the peak are retained. This radius also depends on the SNR, allowing the cluster size to expand in high-SNR conditions and shrink when the SNR is low.

If no bins satisfy the threshold condition, the algorithm falls back to a default window centered around the strongest peak.

Finally, the algorithm determines the smallest delay window that covers all selected bins. Since the delay-domain representation is circular, a wrap-around handling technique is used to correctly determine the minimal span. The resulting window length and starting position are then used to generate the delay-domain window that filters the channel estimate.

This adaptive approach allows the window to automatically adjust to the energy distribution of the CIR and the current SNR, ensuring that the dominant multipath components are preserved while minimizing the inclusion of noise.

Additional remark for high-SNR and noiseless conditions.

In very high SNR, noise suppression is no longer the only design criterion. Although a lower threshold tends to enlarge the selected delay window, the window must still remain smaller than the cyclic shift spacing between pilots. Otherwise, the window may begin to capture neighboring cyclically shifted pilot replicas, which introduces interference into the channel estimate. Therefore, in the final implementation the adaptive window is additionally constrained by the pilot shift spacing, so that the maximum safe window length is limited to approximately shift_step - 5 delay bins.

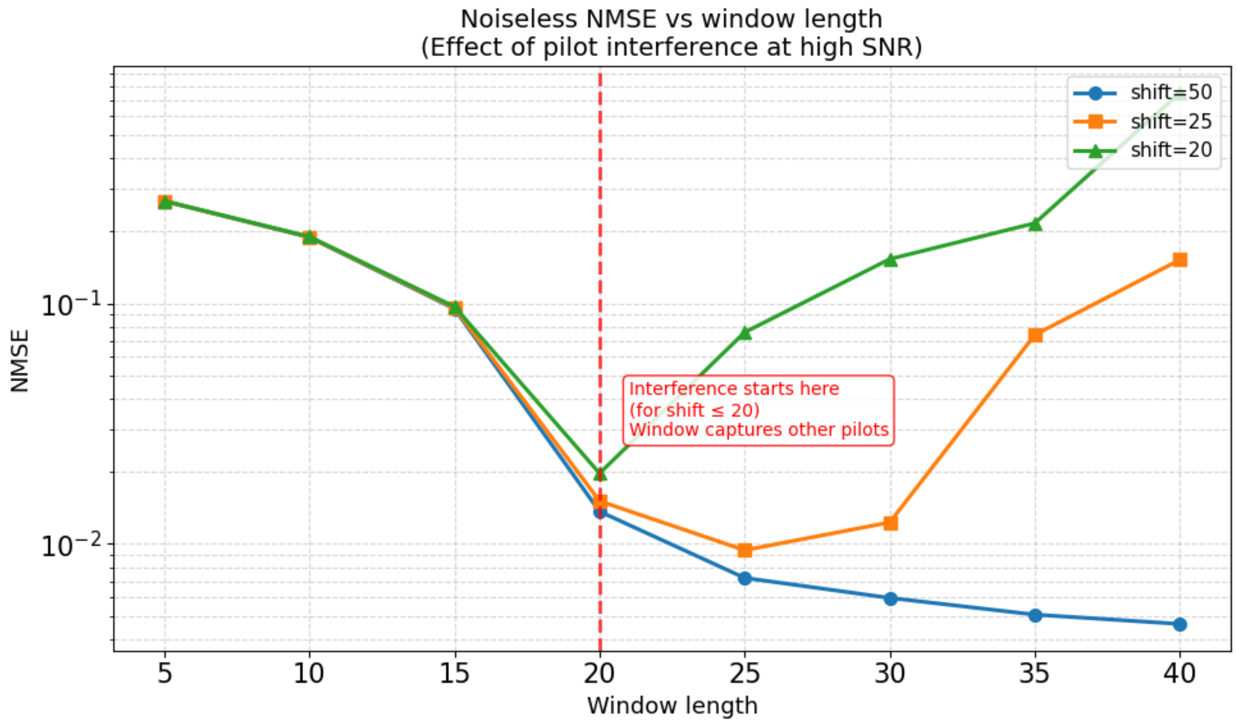
This means that the adaptive window remains data-driven and SNR-dependent, but it is also protected against overlap with neighboring pilots in high-SNR or noiseless cases.

3.1 High-SNR / Noiseless Window-Length Tradeoff and Pilot Interference

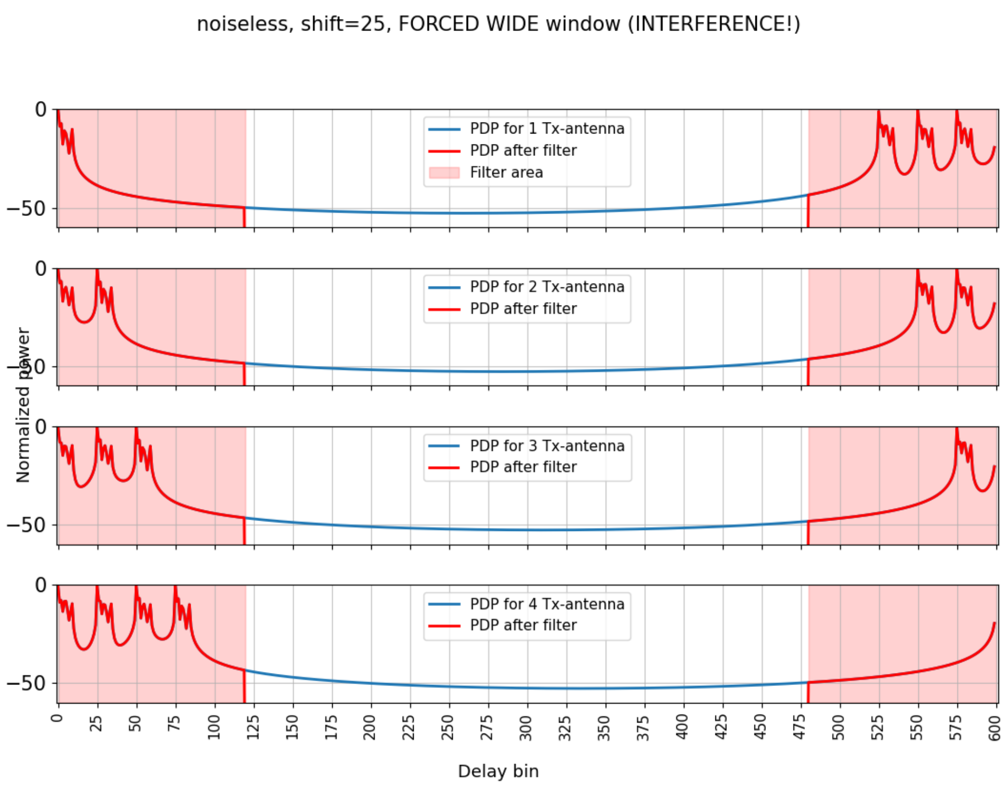
To investigate the windowing response in high SNR conditions, I additionally considered the high-SNR/noiseless regime and compared different cyclic shift spacings. The main aim to see whether the delay-domain window should always be made as wide as possible when noise is negligible.

The results show that this is **not** the case. For a large cyclic shift spacing (shift = 50), the estimation error continues to decrease or remains stable as the window becomes wider, because the neighboring pilot replicas are still sufficiently separated in the delay domain. However, when the cyclic shift spacing is reduced (shift = 25 and shift = 20), the NMSE begins to increase once the window length becomes comparable to the shift spacing.

This happens because the delay-domain window starts capturing energy not only from the desired pilot response, but also from neighboring cyclically shifted pilot replicas. Therefore, in the high-SNR or noiseless regime, the optimal window is not the widest possible one. Instead, it must be limited by both the channel delay spread and the cyclic shift spacing.



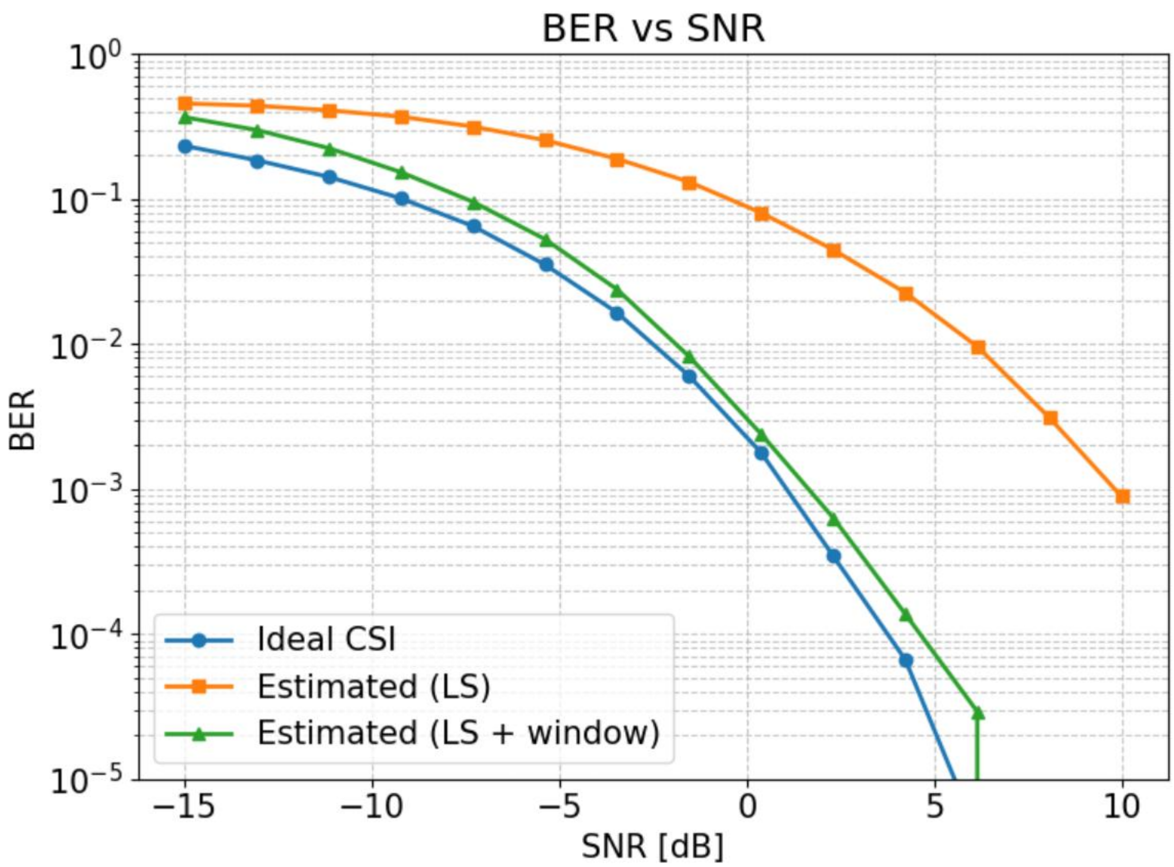
The NMSE-versus-window-length plot confirms this effect: for small shifts, the curve first decreases as the useful channel energy is captured, but then increases again once pilot interference enters the window.



4. BER vs SNR results and performance gap analysis

The BER vs SNR curves compare three different cases:

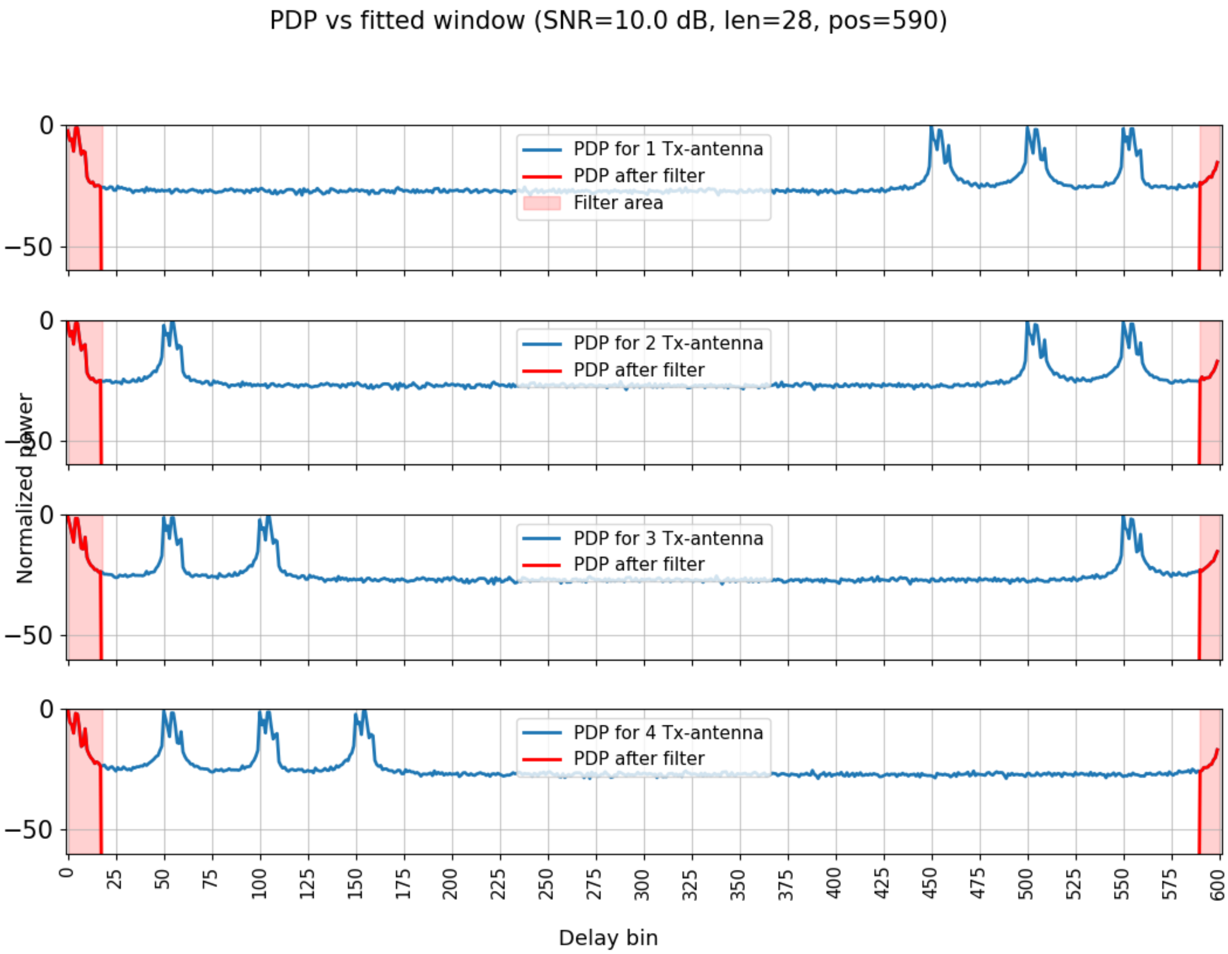
- **Ideal CSI** precoder designed using the true channel
- **Estimated (LS)** precoder designed using the noisy LS channel estimate
- **Estimated (LS + window)** precoder designed using the filtered channel estimate



From the results, several observations can be made.

- First, the **ideal CSI curve achieves the best performance**, since the transmitter has perfect knowledge of the channel and can align the precoder exactly with the dominant singular mode of the channel.
- Second, the **LS estimated channel performs significantly worse**, especially at medium and high SNR values. This is because the LS estimate contains substantial noise, which distorts the estimated channel matrix. As a result, the singular vectors used for precoding are not perfectly aligned with the true channel, reducing beamforming gain and increasing the bit error rate.

- Third, applying **delay-domain filtering (LS + window)** significantly improves the performance compared to the raw LS estimate. By removing noise-dominated delay bins, the filtered channel estimate more accurately represents the true multipath structure of the channel. Consequently, the precoder designed from the filtered channel is closer to the optimal one.



4. Control Questions:

1. What properties of Zadoff-Chu sequencies make them suitable for channel estimation?

Zadoff–Chu sequences are well suited for channel estimation because they have several important properties.

First, they have constant amplitude, meaning the magnitude of all sequence elements is the same. This results in a **low peak-to-average power ratio (PAPR)**, which is beneficial for transmission since it avoids large signal peaks that could distort the signal in power amplifiers.

Second, Zadoff–Chu sequences have **ideal cyclic autocorrelation properties**. Their cyclic autocorrelation is zero for all non-zero shifts and has a strong peak at zero shift. This property allows the receiver to clearly detect the timing and estimate the channel by correlating the received signal with the known sequence.

Third, different Zadoff–Chu sequences (with different roots or cyclic shifts) have **low cross-correlation with each other**, which allows multiple users or antennas to transmit pilot signals simultaneously while still being separable at the receiver.

Because of these properties Zadoff–Chu sequences are widely used in communication systems for pilot transmission and channel estimation.

2. What are four common methods of orthogonal multiplexing using Zadoff-Chu sequencies?

Here are four common methods of orthogonal multiplexing using Zadoff-Chu (ZC) sequences.

1. **Cyclic Shift Orthogonality (CS)** Different transmit antennas or users apply different cyclic shifts to the same root ZC sequence. Due to perfect cyclic autocorrelation, shifted versions are orthogonal in the delay domain (correlation = 0 for non-zero shifts).

Used for multi-antenna SRS (Sounding Reference Signal).

2. **Different Root Indices (Different Roots):** Different users or cells use ZC sequences with different root indices r. Cross-correlation between different-root ZC sequences has constant low magnitude $|R(\tau)| = 1/\sqrt{K}$ for all τ.

Enables multiple users to transmit pilots on the same time-frequency resources with low interference.

3. **Time-Domain Orthogonal Cover Codes:** (OCC) ZC sequences (or their cyclic shifts) are multiplied element-wise by short orthogonal cover sequences (e.g., length-2: [+1, +1] and [+1, −1]) across consecutive OFDM symbols or slots.

Provides additional orthogonality in time domain, increasing the number of orthogonal pilots.

4. **Frequency-Domain Orthogonal Multiplexing**: Different users/antennas are assigned interleaved subcarriers (comb pattern), and each comb is modulated by a ZC sequence (or its shift/root).

Combined with cyclic shifts or OCC, this creates orthogonality in frequency domain.

These four methods (cyclic shift, root separation, OCC, and frequency comb) are the most common ways ZC sequences achieve orthogonality for multi-user/multi-antenna channel estimation and reference signals in modern wireless systems.

3. **How does the cyclic shift parameter affect the delay-domain separation between pilot signals?**

The cyclic shift parameter determines the **delay-domain separation** between pilot signals from different antennas or users.

A cyclic shift of L samples applied to the base Zadoff-Chu sequence (via a phase ramp $\exp(j\ 2\pi\ k\ L\ /\ K)$ in frequency domain) shifts the correlation peak of that pilot's channel response exactly by L delay bins at the receiver.

Due to the perfect cyclic autocorrelation of ZC sequences, shifted versions remain orthogonal (zero cross-correlation for non-zero shifts), allowing clean separation of channels via delay-domain windowing around each distinct peak.

In this implementation, the cyclic shift step of 50 bins ensures sufficient separation for channels with short delay spread, preventing overlap and enabling accurate multi-antenna estimation on shared resources.

Additional simulations with smaller cyclic shift spacings (shift = 25 and shift = 20) showed that overlap risk becomes visible much earlier, confirming that the maximum safe delay-window length is strongly constrained by the cyclic shift spacing.

4. Why does delay-domain filtering improve channel estimation accuracy?

Delay-domain filtering improves accuracy by exploiting the fact that real channel multipath energy is concentrated in a limited number of delay bins (short delay spread), while AWGN noise from LS estimation is uniformly spread across all delay bins.

Applying a mask that keeps only the main cluster (signal-dominated) and suppresses the rest (noise-dominated) removes a large portion of estimation noise without losing significant channel information.

After filtering in delay domain and transforming back to frequency domain, the resulting channel estimate has lower noise variance, leading to improved precoding performance and reduced BER, as shown in the simulation results. The adaptive window length further optimizes this trade-off across SNR levels.

5. What is the physical meaning of power-delay profile?

The power-delay profile (PDP) represents the distribution of received signal power as a function of propagation delay τ.

Physically, it describes how multipath components (direct path and reflections) arrive at the receiver at different times due to varying path lengths. Each peak in the PDP corresponds to a distinct multipath component, with its height indicating the power (strength) of that path and its position indicating the relative delay.

The PDP quantifies the time dispersion of the wireless channel, revealing the delay spread and multipath structure essential for understanding channel selectivity,and designing effective equalization or filtering strategies. In the simulation, the PDP shows very short delay spread with energy concentrated near $\tau \approx 0$ and wrapped around $\tau \approx 600$ due to circular IDFT properties.

6. Define delay spread. How can it be estimated from the PDP?

Delay spread quantifies the time dispersion of a wireless channel due to multipath propagation it measures how widely the received signal power is distributed across different arrival delays.

It is formally defined as the **root mean square (RMS) delay spread** σ_τ, which is the standard deviation of the normalized power delay profile (PDP):

1. Normalize PDP: $p(\tau) = PDP(\tau) / \sum PDP(\tau)$
2. Mean delay: $\mu = \sum \tau \cdot p(\tau)$
3. Second moment: $\mu_2 = \sum \tau^2 \cdot p(\tau)$
4. RMS delay spread: $\sigma_\tau = \sqrt{(\mu_2 - \mu^2)}$

This value is estimated directly from the PDP by computing the power-weighted variance of delay τ, giving the effective duration over which significant multipath energy arrives. In the simulation, RMS delay spread is calculated in bins for both channels, reflecting their short dispersion.

7. How does delay spread affect communication system perfomace?

Delay spread (RMS delay spread σ_τ) quantifies the time dispersion of multipath components in a wireless channel. Larger delay spread leads to:

- **Increased frequency selectivity** deeper fades across subcarriers, reducing effective SNR in some frequencies.
- **Higher risk of inter-symbol interference (ISI)** echoes from previous symbols overlaps with the current one if delay spread exceeds symbol duration.
- **Greater equalization complexity** requires longer cyclic prefix in OFDM or more equalizer taps.
- **Higher outage probability and lower reliability** deep fades can cause signal drops, especially at low SNR.

Small delay spread (as observed in both channels) results in limited dispersion, simpler equalization, and better performance.

In contrast, large delay spread degrades throughput, increases error rates, and requires more advanced mitigation techniques such as longer cyclic prefix, MIMO diversity, or adaptive modulation.

8. Why should pilot overhead be minimized in practical communication systems?

Pilot overhead refers to the fraction of time-frequency resources used for transmitting reference signals instead of data. It should be minimized because:

- It reduces **spectral efficiency** and **data throughput** resources spent on pilots cannot carry user data.
- It increases **latency** and **power consumption** pilots require transmission time and energy.
- It limits **multi-user capacity** excessive pilots cause contention and interference (pilot contamination) in dense networks.

Excessive overhead lowers overall system performance, while too few pilots degrade channel estimation accuracy.

9. If pilot symbols are transmitted only on a subset of subcarriers, what additional processing is required to estimate the full channel?

When pilots are transmitted only on a subset of subcarriers (e.g., comb pattern), channel estimates are available only at pilot locations. To obtain the full channel response across all subcarriers, frequency-domain interpolation is required.

Common methods include:

- **Linear or polynomial interpolation** between pilot estimates.
- **Transform-domain interpolation**: IFFT pilot estimates to delay domain, apply windowing/truncation to keep significant paths, FFT back to frequency domain for all subcarriers.
- **Sinc or MMSE interpolation** using channel statistics.

This additional processing reconstructs the frequency-selective channel response, compensating for the sparse pilot grid while balancing estimation accuracy and pilot overhead. In dense pilot cases (as in this simulation), no interpolation is needed.

10. Can multiple users estimate their channels simultaneously using the same time–frequency resources? Under what conditions?

Yes, multiple users can estimate their channels simultaneously using the same time–frequency resources, provided the pilot sequences assigned to them are orthogonal or have sufficiently low cross-correlation.

Key conditions (using Zadoff-Chu sequences):

1. **Different root indices** Users are assigned ZC sequences with different roots $r_1 \neq r_2$. Cross-correlation magnitude between different-root sequences is constant and low: $|R(\tau)| = 1/\sqrt{K}$ for all τ, minimal interference.
2. **Orthogonal cyclic shifts** (within the same root) Within a cell or group, users/antennas use the same root but different cyclic shifts. Perfect cyclic autocorrelation ensures orthogonality: correlation is zero for non-zero shifts, provided the shift spacing exceeds the channel delay spread (to avoid overlap).
3. **Sufficient shift spacing** Cyclic shift step must be larger than the maximum expected delay spread of the channel, prevents overlap of correlation peaks in delay domain.

11. How does channel estimation errors affect precoder performance?

Channel estimation errors occur when the base station computes the precoder W using an imperfect estimate of the downlink channel H_dl instead of the true channel. These errors, primarily from noise in uplink pilots, lead to:

- **Beam misalignment** the precoder vector deviates from the optimal dominant singular direction, reducing the array gain and effective received signal strength at the UE.
- **Lower effective SNR** the beamformed channel gain $||H\_dl\ @\ W||^2$ is smaller than with perfect CSI, increasing bit error rates.
- **Increased residual interference** in multi-user or non-ideal scenarios, misaligned beams leak energy to unintended directions, degrading SINR.

In the simulation, raw LS estimation (noisy CSI) produces a noticeable BER gap compared to ideal CSI (blue curve). Delay-domain windowing reduces estimation errors by suppressing noise outside the main multipath cluster, resulting in a cleaner channel estimate, more accurate precoder, and significantly lower BER (green curve closer to blue), demonstrating that better CSI quality directly enhances precoding performance and overall reliability.

12. What would happen if the filtering window is chosen too small or too large?

If the filtering window is too small, some true multipath components are excluded from the estimate. As a result, part of the useful channel energy is lost, which increases estimation error and degrades the precoder quality.

If the filtering window is too large, two different effects may occur depending on the SNR. At low SNR, the window includes many noise-dominated delay bins, which increases estimation noise. At high SNR or in the noiseless case, the more important problem is that the window may start to capture neighboring cyclically shifted pilot replicas. This introduces pilot interference into the channel estimate and may increase NMSE even though the noise level is very low.

Therefore, the window length must be chosen as a tradeoff: it should be wide enough to contain the dominant channel energy, but not so wide that it includes either excessive noise or neighboring shifted pilots.

13. Can precoding be applied during pilot transmission? How would it affect channel estimation accuracy and system design?

Precoding can be applied during pilot transmission, but it is not the standard approach in most TDD massive MIMO systems (including 5G NR SRS), where pilots are transmitted unprecoded.

If precoding is applied to pilots (Y = H @ (W_pilot @ S) + noise):

- Channel estimation accuracy is degraded, the BS estimates the effective channel H @ W_pilot instead of the true H, leading to biased, incomplete, or directional estimates. This distorts the channel observation and reduces estimation quality, especially if W_pilot is outdated or mismatched.
- System design becomes more complex, requires prior knowledge or feedback of W_pilot, increases overhead, limits multi-user flexibility, and complicates adaptive precoding. It may be used in specific scenarios, but it reduces the BS's ability to obtain a full, unbiased channel view.

In this simulation, pilots are transmitted unprecoded, enabling accurate estimation at the BS, followed by windowing and precoding only for data transmission.